

Automated and Monitoring Systems Based on IoT for Smart Poultry Farm Management: A Review

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Abstract: *IoT technologies have fully transformed traditional chicken farming into intelligent and highly automated systems through their integration and activation. The paper looks at how well IoT-based automated monitoring systems manage smart chicken farms. This part emphasizes essential tools coupled with scholarly contributions and practical uses of these technologies to raise operational efficiency and production, thus advancing greater animal care. The study looks at problems with money and putting systems together, especially ones that make it hard to trust the data. The idea lets you use blockchain technology along with good data management techniques to make data more secure and easier to track. IoT solutions assist poultry farms in monitoring and modifying crucial environmental and operational factors, thereby ensuring optimal conditions for animal care and production. IoT-based solutions help farms to save running costs and simultaneously increase output and animal welfare. Ensuring data accuracy and system integration expenses are still the key challenges for applications. This article offers IoT technology to improve safety and openness, as well as remedies for these problems via thorough environmental monitoring.*

Keywords: *Internet of Things, automated systems, poultry farm management, real-time monitoring, operational efficiency, environmental parameters.*

I. INTRODUCTION

A. Background on Poultry Farming

Since the advent of agriculture, birds have been domesticated for their meat, bones, eggs, and even feathers. This practice is known as poultry farming, one of the most strategic types of animal husbandry, which makes use of waste products from food production, domestic farming, agriculture, and even industrial waste. For many people living in rural areas all over, especially in developing countries, it provides a consistent source of protein in addition to food and nutritional stability.[1]. Apart from these low-impact home production approaches, poultry is also one of the most commonly used animal species in industrial livestock farming. Most commercial chicken farming is indeed industrial poultry farming, mostly from chicken and, to a lesser degree, from turkey, duck, geese, and other species, which derive the large majority (up to 98% for meat and 92% for eggs) of the world's poultry output. [2]. By 2021, global chicken meat output is projected to rise to 137.8 million metric tons. [3].

B. Environmental and human impact

Organic, free-range, lower-density, and low-impact agricultural methods are growing in the EU to enhance farming, animal welfare, environmental impact, and sustainability. They employ slower-

growing chicken genotypes, fewer animals with outside grazing areas, and more locally produced feed. [4] With over 12% annual growth[5], about 10% of EU chicken is grown less intensively. Integrating locally produced feeds, especially poultry, into the agricultural system and using marginal lands can improve sustainability in large-scale production.

C. Profitability and Risk

Poultry farming is profitable because people desire chicken and eggs. Many industry efforts have failed, making them risky. Poor animal care, money management, and disease outbreaks create huge losses. Chicken care is important because the company is complex and the market changes often. Hard work and few resources make traditional farming dangerous. To be lucrative and efficient, poultry farming must use advanced technologies due to increased livestock demand. [6][7].

D. Critical Environmental Factors

Temperature, ammonia levels, and food availability are things that are often ignored in chicken farming. These things have an impact on birds. For instance, ammonia levels in the cage can cause respiratory and psychological problems for the birds. Extreme temperatures hamper the growth and reproduction of birds. An inadequate supply of food will have an impact on the bird's productivity. To improve chicken quality and quantity, poultry farmers must solve these ecological issues.[8] [9]. Networked gadgets have changed poultry farming. The Internet of Things makes real-time feeding, humidity, and temperature management possible. This optimal atmosphere keeps birds healthy and prolific. IoT technology must be employed more in farming to save resources, increase efficiency, and reduce environmental impact.[10] [11].The Role of IoT in Poultry Farming.

E. Benefits of IoT-Driven Systems

- **Animal Welfare and Operational Efficiency:** New studies show that systems built on the Internet of Things (IoT) have the potential to boost animal welfare, increase operational efficiency, and decrease operating costs [12]. Poultry farmers can gain access to precise data in real-time through the use of IoT technologies, allowing them to make educated decisions and treatments right away. The development of thorough monitoring systems that will enable farmers to keep an eye on things and quickly make modifications has also been greatly aided by innovative sensor technology. Therefore, the Internet of Things (IoT) causes a proportional rise in production and enhances standards for animal welfare by making sure that the chickens live in better circumstances [13].
- **Cost-effective Management Systems** are made possible by letting farmers remotely monitor system operations, and deployable IoT management systems are

transforming chicken production via web apps. In less developed countries, such systems may assist with problems like food and nutrition. These qualities provide control of temperature, humidity, water supply, and ammonia levels, therefore improving resource management and reducing labour expenses [14]. These systems also include biosecurity protections, automated features allowing farmers to simply maintain perfect conditions with less direct oversight, and individualized light scheduling.

- **Enhancing the Quality of Control and Monitoring:** Research studies show that IoT platform technology helps control and monitor important environmental factors for best poultry farming[15]. Through sensors and data analysis, farmers can maintain a suitable environment for chickens, including temperature, humidity, and ammonia levels. Efficient control of these elements[16] drives animal growth, feed conversion efficiency, and overall health. By employing IoT technologies, farmers can quickly respond to environmental changes, ensuring crop quality and controlling the welfare of livestock [17].

II. LITERATURE REVIEW

Research in the past several years demonstrates the potential of IoT in agriculture, especially in smart farming. Some crucial papers on this matter are outlined below.

The brand-new traits and packages of IoT in clever rooster farming are reviewed, covering numerous aspects of IoT deployment, which include sensor networks, statistics analytics, and automated manipulation systems. [18] The effort enhances ecological, individual, and operational parameters through the use of detectors, data analysis, and computerized controls.

Contemporary technologies enhance poultry growth, feed efficiency, and agricultural management. [19]. An extensive examination of poultry IoT applications explores innovative agriculture technology to improve productivity. Sensors, data analytics, and automated controls optimise conditions, animal health, and efficiency. Growth rates, feed conversion ratios, and farm management improved significantly. [20]. A comprehensive 2020 Poultry Tech Summit evaluation of IoT solutions in smart farming showed the revolutionary power of sentence-level representations. These enhanced representations capture subtle meanings, improving translation and summarization. Efficiency and accuracy gains suggest more innovative, more sustainable farming.[21].

The intersection between IoT technologies and machine learning in smart poultry farming is discussed, namely, how decisions using predictive analytics can provide foresight to enhance management performance and undertakings. [22]. Invest mainly focuses on the role of IoT in improving resource management in poultry farms, particularly in monitoring and management of water and feed, the distribution of these resources, and the economic and environmental consequences of adopting IoT technology. [23]. Other IoT-based solutions for the health monitoring of poultry include the use of wearable sensors and biometric devices for the early detection and intervention of health issues in poultry.

An elaborate IoT framework for smart poultry farming is presented, which utilizes diverse sensors, communications frameworks, and processing units to deliver a scalable and interoperable system. The design and implementation of an automated chicken feeder for chicken farms was studied, focusing on the positive

effects automation has on labor costs and feeding efficiency [24]. A decision-poultry farm management system is developed based on IoT and real-time data analysis to provide analytics and empower better decision-making. [25].

The integration of IoT and AI in smart poultry farming demonstrates how AI algorithms can analyze data collected by IoT devices to provide predictive insights and recommendations. This innovative merging of AI and IoT in poultry farming highlights how AI algorithms can manage large amounts of data from IoT devices for real-time monitoring and predictive insights. This approach improves efficiency and productivity, ensuring the well-being of the poultry by enabling timely, data-driven decisions. It is undoubtedly a massive advancement for the sector [26].

IoT-based systems can be used for predictive maintenance on poultry farms. These systems can monitor equipment performance and predict potential faults, which helps reduce downtime and improve operational efficiency. The Conference on Smart Instrumentation, Measurement, and Application highlights the development of a smart chicken poultry farm using a Real-Time Operating System on Arduino. The research aims to automate poultry management to boost efficiency. The text presents a collaborative investigation and performance to show how the equipment works. It will be a valuable material for those fascinated by the new agricultural technology [27].

This creative idea is aimed at improving agricultural management by employing the IoT to extract current temperature, humidity, and air and bird health status data. This Internet of Things (IoT) system shows significant variations in management practices accompanying agricultural production and operational efficiency. The main benefits of the system are improved resource allocation, reduced labor costs, and scalable chicken production techniques. By means of automation of repetitive chores, the system guarantees the best conditions for poultry [28] and supports profitable, sustainable farming techniques. An IoT-based smart poultry farming system with built-in automation is described, stressing automation's advantages in enhancing agricultural efficiency and animal welfare[29]. The design of a sensor-based IoT system intended to boost poultry performance and seed production is described, stressing the need for real-time monitoring, data analysis, delivery, and passive management [30].

IoT-based environmental monitoring systems have been identified for poultry farms, focusing on their efficacy under optimal conditions for poultry health and seed production[31].

Smart farming uses IoT data analytics and real-time monitoring to improve agricultural efficiency and yield. The article focuses on how IoT-enabled predictive analytics may improve chicken farm decision-making and proactive management. It discusses IoT-based poultry farming systems, including their key benefits and the need for system data tracking to make judgements [32]. The integration of IoT and wireless sensor networks in smart poultry farming is discussed, citing the benefits of WSNs in offering real-time data and supporting computerized control systems[33].

After the researcher focused on using the sensor network and mobile devices to manage and monitor climatic and agricultural restrictions, we can move on to wireless sensor networks. Wireless monitoring devices send mobile phone messages to owners about temperature and moisture changes. The system can reply to the user by commanding the control system to increase or decrease fan speed to comfort the birds. With a low-cost innovative poultry monitoring system, the farm was controlled[34].

How to make an inexpensive, standardized environmental monitoring device with the Raspberry Pi (R-Pi) CPU. A Python-based system can be administered remotely via IoT. Ecological data from sensors is accessible 24/7 online. It detected temperature, humidity, light intensity, and carbon monoxide accurately in trials[35].

Another writer proposed ZigBee-based online chicken farm monitoring. Monitoring in real time will be networked. System components under monitoring are node controllers, data receivers, and ZigBee control nodes. TGS4161 CO2 and SHT75 temperature and humidity sensors measure ambient conditions. Analysis of system data transfer and ZigBee protocol stack simplification will generate data transmission methods and communication formats [36]. We propose a wireless sensor network (WSN) to measure poultry ambient parameters such as temperature, humidity, air quality, and light intensity. Many sensors are carefully placed across the property for more precise data. ZigBee transceivers sent aggregated agricultural data to a PIC microcontroller or PLC module-based primary controller via a wireless sensor network.

Also controlled is the water level [37]. Have an automatic sprayer, smart lighting, and farm temperature management with sensitive motion sensors that work when needed. System energy savings were efficient. [38] Introduced a humidity table-connected method to

monitor bird health factors like heart pulse, skin temperature, and stress, as well as climate and farm variables like humidity. In chicken farms, temperature, humidity, water temperature, and valve control are measured [39], [40]. Remote sensors send values to the server, which can be seen on Google spreadsheets.

Operating an IoT-based system sustainably under off-grid conditions involves utilizing renewable energy sources, minimizing power consumption, and ensuring the longevity of all components. Solar energy is especially beneficial since it offers a consistent, clean power source. Using energy-efficient hardware, recyclable materials, and following eco-friendly design ideas will help attain even more sustainability. The system's sustainability is also enhanced by the capacity to remotely track and control power consumption since problems can be fixed right away without needing physical intervention[41].

Below is Table 1, which summarizes all the related work explained in the literature review.

TABLE.1 SUMMARIZES ALL THE LITERATURE REVIEW

<i>Ref.No.</i>	<i>Methodology</i>	<i>Sensors Used</i>	<i>Microcontroller Used</i>	<i>Problem Highlighted</i>	<i>Specified IoT</i>
[18]	Cloud-Based IoT Solution for Enhanced Poultry Farm Management	Temperature sensors, Humidity sensors, Gas sensors	ESP8266	Challenges in real-time monitoring and management of poultry farms	IoT deployment involving sensor networks, data analytics, and automated control systems
[19]	IoT-based smart farming automation system leveraging data analytics and automated controls	Temperature sensors, Humidity sensors, CO2 sensors, Light sensors	Arduino Uno	Environmental condition optimization and livestock health monitoring	Real-time data collecting and analysis using an IoT platform
[20]	Data analytics and automated control systems	Temperature sensors, humidity sensors, and CO2 sensors	ESP32	Optimization of poultry farming conditions	IoT-based smart farming system
[21]	Systematic Review	Temperature, Humidity, Soil Moisture, Light Sensors	Arduino, Raspberry Pi	Inefficient, traditional farming practices	Smart farming using IoT to enhance resource use and crop productivity
[23]	Optimal resource management methodology for IoT-based green and sustainable smart cities	Temperature sensor, humidity sensor, light sensor	Arduino Uno	Challenges in managing resources efficiently in IoT-based smart cities	Green and sustainable IoT for smart cities
[24]	IoT-based real-time monitoring	Temperature and humidity sensors	ESP8266	Health status identification of poultry	IoT-based solutions for poultry farming
[25]	A comprehensive IoT framework for smart poultry farming, integrating various sensors, communication protocols, and data processing units.	Temperature sensors, humidity sensors, and feed level sensors	Arduino Uno	High labor costs and inefficient feeding processes in poultry farms.	Interoperable and scalable IoT solution for poultry farming.
[26]	Real-time data analytics and monitoring system	Environmental sensors (temperature, humidity, etc.)	Arduino	Improvement of poultry farm management efficiency	Internet of Things (IoT) for real-time monitoring and decision-making
[27]	Implementation of smart sensors, big data analysis, and IoT technologies in poultry management	Temperature, humidity, CO2, and ammonia sensors	ESP8266	Environmental control challenges and animal health monitoring inefficiencies	IoT-based real-time monitoring system
[28]	Implementation of RTOS on Arduino for real-time monitoring and control of poultry farm operations	Temperature sensors, Humidity sensors, Gas sensors	Arduino	There is a need for automation and real-time monitoring in poultry farms to improve efficiency and reduce manual labor.	IoT-based system for predictive maintenance and environmental monitoring
[29]	IoT-based automation system for real-time monitoring of environmental conditions and chicken health indicators.	Temperature, humidity, and air quality sensors	Arduino	Need for efficient and scalable poultry farming solutions	Use of IoT for farm management and automation
[30]	Development of a smart chicken poultry farm with integrated automation	Temperature sensors, humidity sensors, and gas sensors	Arduino Mega 2560	Improving agricultural efficiency and animal welfare through automation	IoT-based intelligent poultry farming system
[31]	Automated techniques for monitoring the behavior and welfare of broilers and	Various sensors, including cameras and microphones	N/A	Improving precision livestock farming by monitoring behavior and welfare	IoT-based real-time monitoring system

Ref.No.	Methodology	Sensors Used	Microcontroller Used	Problem Highlighted	Specified IoT
	laying hens				
[32]	Smart Poultry Farm Automation and Monitoring System	Temperature, Humidity, Gas Concentration Sensors	Arduino	Efficient and Real-time automation in poultry farms	IoT-Based Monitoring System
[33]	Review of IoT-based data management methods in smart farming	Environmental sensors (temperature, humidity), motion sensors	ESP32 microcontroller	Complexity in data integration and real-time monitoring	IoT-enabled predictive analytics and real-time monitoring systems
[34]	Smart Poultry Farm Monitoring Using IoT and Wireless Sensor Networks	Temperature, humidity, gas, and light sensors	Arduino Uno	Lack of real-time data and manual labor in poultry farms	IoT-enabled monitoring system
[35]	The combined approach of sensor networks and mobile devices	Temperature, moisture sensors	N/A	Climate condition management and monitoring	Wireless sensor networks
[36]	Cost-effective, standardized environmental monitoring devices using R-Pi and Python.	Temperature, humidity, light level, and CO sensors	Raspberry Pi	Accurately measure environmental parameters and upload data to the internet.	IoT platform
[37]	Automated control system for poultry farms using embedded systems.	Temperature, Humidity, and Ammonia gas sensors	Arduino ATmega2560	Monitor and control environmental parameters automatically	SMS gateway service
[38]	Automatic sprayer with a smart lighting system and control of the temperature of the farm using motion-sensitive sensors.	Motion Sensors, Temperature Sensors	N/A	Efficient energy saving in poultry farms.	IoT technology for smart farm management
[39]	Mobile system network with wireless sensors for climate control	Wireless sensors	N/A	Remote monitoring of the birdhouse climatic conditions	Zigbee-Based Health Monitoring System
[40]	IoT-based Smart Management of Poultry Farms and Electricity Generation	Temperature, Humidity, CO2 Sensors	ESP8266	Poultry waste management for a biogas power station	Integration with Google Spreadsheets for data viewing
[41]	A comprehensive literature review of renewable energy technologies applied in poultry farming	Temperature sensors, Humidity sensors	Arduino	emphasises that chicken production is energy-intensive, requiring a lot of fossil fuel to maintain interior climatic conditions, which raises operating costs and GHG emissions.	IoT technology for smart farm management

III. PROPOSED SYSTEM

The introduction plan for using IoT technology in rearing chickens is a progress in agribusiness. A complicated system of sensors will be introduced in a chicken coop to detect the levels of humidity, temperature, ventilation, and the threat of fire. The data will be sent to a single control center, where it will be processed to identify the problem rapidly and inform the farm manager about the necessity of taking certain measures. The approach will contribute to the reliability of the whole system due to the possibility of preventing potential threats to the environment and the farm. The introduction of technological innovations can increase the quality and sustainability of chicken farming by controlling the environment, establishing systems of health management, and economizing the use of resources to reduce waste and enhance productivity.

The anticipated uptake of the Internet of Things (IoT) framework in chicken farming facilitates a huge leap in agricultural oversight. This system relies on a set of detectors strategically located in a chicken house as a measure of monitoring various environmental conditions, including humidity levels, temperature, air quality, and risks of fire threats. The data gathered is used to immediately trigger responses in case of any unwanted situation, which are analyzed from a central location. The detection technology used to monitor these situations is sophisticated, with automated alarm systems. This proactive approach enhances the reliability of the system by pointing out risks before they come to fruition in production environments.

A. Automation and Efficiency

This system automates the feeding, watering, and environmental control process, which is used to ensure effective resource management, reduce human labour, and improve the accuracy in resource allocation. One way through which the system maintains the safety of birds and infrastructure is through improved methods of detecting fires and maintaining them properly. By avoiding these simple tasks, the level of efficiency is improved in relation to the operations and activities that are usually associated with the poultry industry.

B. Technological Integration and Benefits

This particular system boasts an advanced technology that enhances the everyday management of a chicken farm and lays the groundwork for future agricultural advancements. It works by offering a technology-based approach to farm management with the help of the Internet of Things (IoT). The technology enables farmers to make informed decisions by providing real-time information on chickens, which in turn, improves the health and growth of the fowls, resulting in the production of high-quality products.

C. Expected Outcomes

Reducing efficiency and output while increasing operating costs is a potential downside to this method. Better chicken care and more sustainable agriculture represent two benefits of incorporating automation and IoT technology into the process. The system updates traditional farming methods with 21st-century technology, ensuring a more environmentally friendly and efficient future for chicken farming.

The components of this smart poultry farm, as illustrated in **Fig.1**, include The components of that smart poultry farm contain the following:

III. A humidity and temperature sensor

A cost-effective temperature and humidity sensor, such as the DHT11, can be used for various applications. Operating through a single-wire interface and featuring a digital output for primary usage, it is easy to integrate into microcontroller-based systems. Its small footprint and low power consumption make the DHT11 ideal for confined spaces and battery-powered devices. Although its accuracy is not the best, its simplicity and low cost make it a common choice for applications where exact readings are not critical, educational purposes, and hobbyist projects. The DHT11 is a simple and affordable option for numerous electronic projects and applications, measuring temperature and humidity.

III.B FIRE SENSOR

Detection and alarm methods of fire are effectively working with smoke detectors, infrared (IR) cameras, and thermal imaging systems. By just looking at the state of heat, smoke, and light, the sensors can easily detect the presence of fire or flames. When the fire or flames are recognized, the fire detection system activates the alarm or extinguishing system, preventing the fire from spreading to the rest of the building or house and informing people living there. To save people's lives and protect property, a fire detection system is essential in many buildings, such as homes, business premises, and factories. Fire sensors are required for a quick reaction to emergencies.

III.C GAS SENSORS

A gas detector measures environmental gas concentrations. Industrial gases, including carbon monoxide, methane, and hydrogen sulfide, are dangerous to human health. The detector also detects propane and natural gas leaks in residences. It will transmit an electrical signal or flash the light to advise inhabitants to take action. Thus, the gas detector monitors the environment, protects workers, and prevents gas-related mishaps.

III.D Heater

Moisture management is needed to maintain appropriate humidity in industrial, agricultural, and interior climate control applications. This arrangement measures humidity using a continuous humidity sensor. When humidity drops below the specified value, the heater evaporates water to raise it. When humidity is low, the heater will boost it. The water may not be excessively dry or evaporate. To regulate moisture, the heater may be turned off or adjusted if humidity exceeds the specified value.

Pseudocode for Heater Control:

```
function controlHeater(humidity) {if (humidity
desiredTemperature) {turnOnFan();} else {turnOffFan(); } }
```

III.E FAN

Sensors recognize temperature ranges and activate fans in temperature monitoring systems. Fans cool and disperse heated air. This technology controls fan speed depending on real-time temperature to improve thermal management, energy efficiency, and product utility. A pump uses an impeller, usually the motor, to push water. The pump draws water in and presses it out using the impeller, pump casing, and engine. Water pumps may be small or heavy and utilized in many industrial applications. It is used for irrigation, surface water drainage, and residential water.

Pseudocode for FAN Control:

```
function controlFan(temperature) {
```

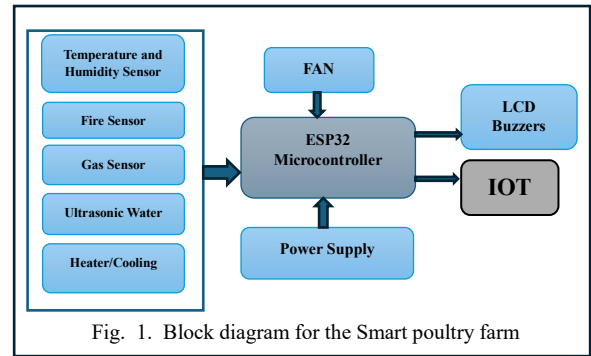
```
if (temperature > desiredTemperature) {turnOnFan();} else {
turnOffFan();}}
```

III. F Liquid Crystal Display (LCD)

Liquid Crystal Display (LCD) technology refers to a flat panel display with two polarizers and a liquid crystal solution sandwiched between them. It consists of a liquid crystal layer sandwiched between two glass sheets. The technology is energy efficient and uses liquid crystals to control the flow of light through them to produce quality images.

III. G Internet of Things

The NodeMCU ESP8266 Wi-Fi module is a modular development board for applications such as the Internet of Things (IoT). By connecting devices to the Blynk app, its user-friendly interface allows users to monitor and control them remotely. Blynk is an intuitive platform with widgets like gauges and buttons linked to virtual or GPIO pins on NodeMCU, permitting users to access sensor data and control actuators in real-time via smartphones or tablets. With minimal programming knowledge, users can develop IoT applications for environmental monitoring and automation using NodeMCU and Blynk.



III. H Components of the Smart Poultry Farm

To enhance comprehension, a summary table is illustrated in Table II:

TABLE.II SUMMARY OF COMPREHENSION

Component	Description
Humidity and Temperature Sensor	The DHT11 sensor measures temperature and humidity and is ideal for small spaces and battery-powered equipment due to its low power consumption.
Fire Sensor	Designed to detect fire or flames by identifying heat, smoke, or light, it triggers an alert or suppression system during fire emergencies.
Gas Sensors	These sensors measure the concentration of gases like carbon monoxide, methane, and hydrogen sulfide to prevent health hazards and accidents.
Heater	Automatically adjusts humidity levels by activating or deactivating water to evaporate in dry conditions or prevent over-evaporation in humid conditions.
Fan	Regulates airflow based on real-time temperature data, dispersing warm air during low temperatures and relieving heat during overheating.
(LCD)	Displays visual information across various devices with

	advantages like compact design, great picture quality, and energy efficiency.
IOT	NodeMCU ESP8266 Wi-Fi module connects devices to the Blynk app for remote monitoring and control, optimizing environmental monitoring and automation.
Buzzers	used to alert the public to any gaps in the environment.

III.I. Addressing Scalability

A scalable system is needed to monitor and regulate environmental parameters across many nodes in big chicken farms. Multiple sensors provide huge data sets that cloud systems can analyze. This enables real-time analysis and decision-making independent of node count. The expandable characteristic of IoT structures boosts their expansion, making it easy to increase the number of nodes without affecting their operations. Other related modes include edge computing. The overload on the central processing unit can be reduced by processing the premature data near the sensors using edge computing. This is essential in either simplifying the process of this data or reducing the latency.

III. J Addressing Multi-node Communication Challenges

For large poultry farms, having good network connectivity is crucial. High-capacity network protocols allow nodes to communicate in real-time for the transmission of information and data. Mesh architecture is inherently self-organizing and scalable, as each node passes on data. This leads to increased robustness and coverage in extensive applications. The synchronization and aggregation of data from different nodes also help to maintain consistency and accuracy. Time-stamps and data aggregated at a central point aid this process.

IV. DISCUSSIONS

After thorough examination and comparison, the device that utilizes NodeMCU code for the IoT integrates with ESP32, providing Blynk connectivity. It has undergone numerous assessments and includes the NodeMCU IoT code alongside the Blynk app and the ESP32. When it was linked with the calibrated instruments or the set parameters, the temperature and humidity readings were obtained, which were accurate. The dependability and responsiveness of the control system in diverse parameter operations according to the sensor input were assessed in varying conditions: heater or fan operations. It was verified that the system could maintain environmental conditions to an appropriate degree of tolerance. From the results illustrated, it was observed that the NodeMCU implementation and Arduino Uno had codes that were appropriate for controlling the sensor readouts and activities. Their nearness to the expected values demonstrated the capability of the devices to make accurate measurements.

Additionally, the ability and promptness of these control systems to respond to variations in external factors such as temperature and humidity resulted in the continuous maintenance of the approved set points for temperature, humidity, etc. The review of the results of the two coding methods shows that NodeMCU and Arduino Uno are the preferred methods for implementing IoT applications with a connection to the Blynk app.

The development and importance of the current system underscore some potential improvements, such as enhancing accuracy through system calibration or more advanced sensors. The system's efficiency and dependability could be considerably improved by altering the software code, for example, by adding error-handling functionality or

changing the control logic processes. The Blynk app's protection, scalability, and UI design were also analyzed, providing potential enhancements and modifications that could cater to consumers' preferences and specifications.

V. CONCLUSIONS

The fusion of IoT technology with the Arduino platform from Blynk has been discovered to be efficient for farmers in managing the hen farm effectively. Due to real-time data acquisition from the sensors, the chicken's living environment and health care are corrected accurately. The implementation of IoT technologies was diverse, resulting in higher efficiency, improvement in productivity, and chicken growth.

The application of IoT technology in poultry farming can make the administration of farms much more efficient and more eco-friendly. The most significant aspect of this, however, is the installation of effective environmental monitoring and control systems in the poultry barns. To check and monitor the humidity and temperature of the barn environment, a system can activate fans and heaters when necessary to control airflow. The system was checked and confirmed after the test, and all the calculations and simulations have been completed. For various reasons, both the NodeMCU and Arduino platforms are suitable for this purpose and have worked better than expected. The user experience on the Blynk App is pretty good, and the user appeared to interact effectively with the system hardware.

This approach makes the system more efficient and user-friendly. When the control and monitoring tasks are automated, it eliminates human intervention while keeping the environment stable and optimal. This saves time and money. Users can access and edit their Blynk settings from any device connected to the internet, which increases the app's usability. Doing things becomes more straightforward, and performance is enhanced. This IoT system improves performance and energy efficiency by streamlining operations. The clients' pleasure is boosted as a result. By increasing the effectiveness and precision of resource management, Blynk encourages intelligent ecological control technologies. This opens the door for future improvements in environmentally friendly technologies.

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